Question	Answer	Marks	Partial Marks
17	Bisector of angle Q accurate with two pairs of correct arcs and Arc centre R, radius 6.5 cm With bird table correctly indicated or implied by correct intersecting constructions	**4**	**M2** for bisector of angle Q accurate with two pairs of correct arcs or **M1** for accurate bisector with no/wrong arcs **M2** for arc centre R, radius 6.5 cm or **M1** for arc centre R Maximum 3 marks if incorrect position/region is labelled, or there is no label and a region is shaded, or *their* constructions do not intersect
18(a)	0.3 oe	**2**	**M1** for 0.4×0.75
18(b)	0.975 oe	**2**	**M1** for $1 - 0.4 \times 0.25 \times 0.25$ oe or $0.6 + 0.4 \times 0.75 + 0.4 \times 0.25 \times 0.75$ or $0.6 +$ *their* **(a)** $+ 0.4 \times 0.25 \times 0.75$
19(a)	$180 - 4x$	**1**	
19(b)	$90 - 2x$	**1**	**FT** *their* **(a)** $\div 2$ in its simplest form dep on expression in x in **(a)**
19(c)	$90 + x$	**2**	**FT** $180 -$ *their* **(b)** $- x$ oe dep on expression in x in **(b)** then fully simplified **M1** for $180 - (90 - 2x + x)$ oe or $180 -$ *their* **(b)** $- x$ oe dep on expression in x in **(b)**
20(a)	$(3y + 2x)(6 - a)$ oe final answer	**2**	**M1** for $3y(6 - a) + 2x(6 - a)$ oe or $6(2x + 3y) - a(2x + 3y)$ oe
20(b)	$3(x + 4y)(x - 4y)$ final answer	**3**	**M2** for $(3x + 12y)(x - 4y)$ or $(3x - 12y)(x + 4y)$ or **M1** for $3(x^2 - 16y^2)$ or for $(x + 4y)(x - 4y)$
21(a)	6	**2**	**B1** for 3^4 or 3^{x-2} or **M1** for $3^x = 81 \times 3^2$ or better
21(b)	8	**3**	**M2** for $x^{\frac{5}{3}} = 32$ or better or **M1** for $\frac{1}{x^{\frac{1}{3}}} = \frac{32}{x^2}$ or better or $x^{-\frac{1}{3}--2} = 32$ or better
22(a)	$\begin{pmatrix} 2 & 17 \\ 10 & -25 \end{pmatrix}$	**2**	**B1** for 2 correct elements

Question	Answer	Marks	Partial Marks
22(b)	2	**2**	**M1** for $-3 - 5k = -13$ oe
22(c)	$\frac{1}{10}\begin{pmatrix} 0 & -2 \\ 5 & 3 \end{pmatrix}$ oe isw	**2**	**M1** for $k\begin{pmatrix} 0 & -2 \\ 5 & 3 \end{pmatrix}$ or for det = 10 or soi
23(a)	Tangent ruled at $t = 24$	**B1**	
	– 0.7 to – 0.3	**B2**	**B2** dep on correct tangent or close attempt at tangent **M1** for rise/run **also dep on** correct tangent drawn or close attempt at tangent. Must see correct or implied calculation from a drawn tangent.
23(b)	acceleration or deceleration oe	**1**	
23(c)	68	**2**	**M1** for $(22 - 5) \times 4$